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Decoding Game: On Minimax Optimality of Heuristic Text Generation Strategies

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TLDR: We proposed Decoding Game and proved that heuristic sampling methods are **near-optimal** strategies, providing **robustness** against **distribution shift** which imposes **implicit regularization** on the objective.

Background: Sampling vs Searching

Open-ended text generation

We want to generate a text $(x_1, \dots, x_T)$ of length T .

We have a language model that computes

$$P(x_1, \dots, x_T) = P(x_1)P(x_2|x_1) \dots P(x_T|x_{<T})$$

Searching	Sampling
Determine $(x_1, \dots, x_T)$ by solving maximum-likelihood $\max_{x_1, \dots, x_T \in V} \log P(x_1, \dots, x_T)$ or its local variants.	Sample next token from a new distribution $Q(x_t) = cP(x_t x_{<t})\mathbf{1}\{x_t \in S\}$ truncation and uniform rescaling.
MAP, greedy, beam, contrastive, ...	top-k, top-p/Nucleus, ...
worse performance	better performance

Sampling outperforms searching in open-ended generation – but why?

Current understanding of this question

- ✗ lacks comprehensive mathematical treatment.
- ✗ does not establish **optimality** results.

Modelling: Decoding Game

Core idea: There is a true distribution $P \neq \hat{P}$ learned by LM.

Assume $TV(\hat{P}, P) \leq \epsilon$. We know **nothing else** about $P \Rightarrow$ **adversarial**.

Two-player zero-sum game:

- Player 1 (strategy)** decides a distribution Q to sample from
- Player 2 (nature)** shifts P within a neighborhood of $\hat{P}$ adversarially
- Payoff** average text quality $\mathbb{E}_{(x_1, \dots, x_T) \sim Q} \log P(x_1, \dots, x_T)$

$$\max_Q \min_{P: TV(P, \hat{P}) \leq \epsilon} \mathbb{E}_{(x_1, \dots, x_T) \sim Q} \log P(x_1, \dots, x_T)$$

Analysis: Minimax Optimality of Truncation-Rescaling Methods

We consider **local** case (single-step generation)

Vectorization $Q, P, \hat{P} \Rightarrow q, p, \hat{p} \in \mathbb{R}^d$

$$\max_q \min_{p: TV(p, \hat{p}) \leq \epsilon} q^\top \log p$$

Theorem 4.3. Define $\hat{w}$ by $\hat{w}_i = \frac{\epsilon}{\log \hat{p}_i - \log(\hat{p}_i - \epsilon)}$. For any given q ,

$$\min_{p: TV(p, \hat{p}) \leq \epsilon} q^\top \log p = q^\top \log \hat{p} - \epsilon \left\| \frac{q}{\hat{w}} \right\|_\infty$$

adversarial objective = non-adversarial objective + ℓ_∞ regularization

Theorem 4.4. Let $\hat{I} = \max \left\{ I: \sum_{i=1}^{I-1} \hat{w}_i \log \frac{\hat{p}_i}{\hat{p}_I} \leq \epsilon \right\}$. Then the optimal strategy

$$q_i^* = c \hat{w}_i \mathbf{1}\{i \leq \hat{I}\} \approx c \hat{p}_i \mathbf{1}\{i \leq \hat{I}\}$$

1st-order uniform tail truncation
approx rescaling

where c is the normalizing constant.

Generally, for a non-decreasing and concave f beyond log,

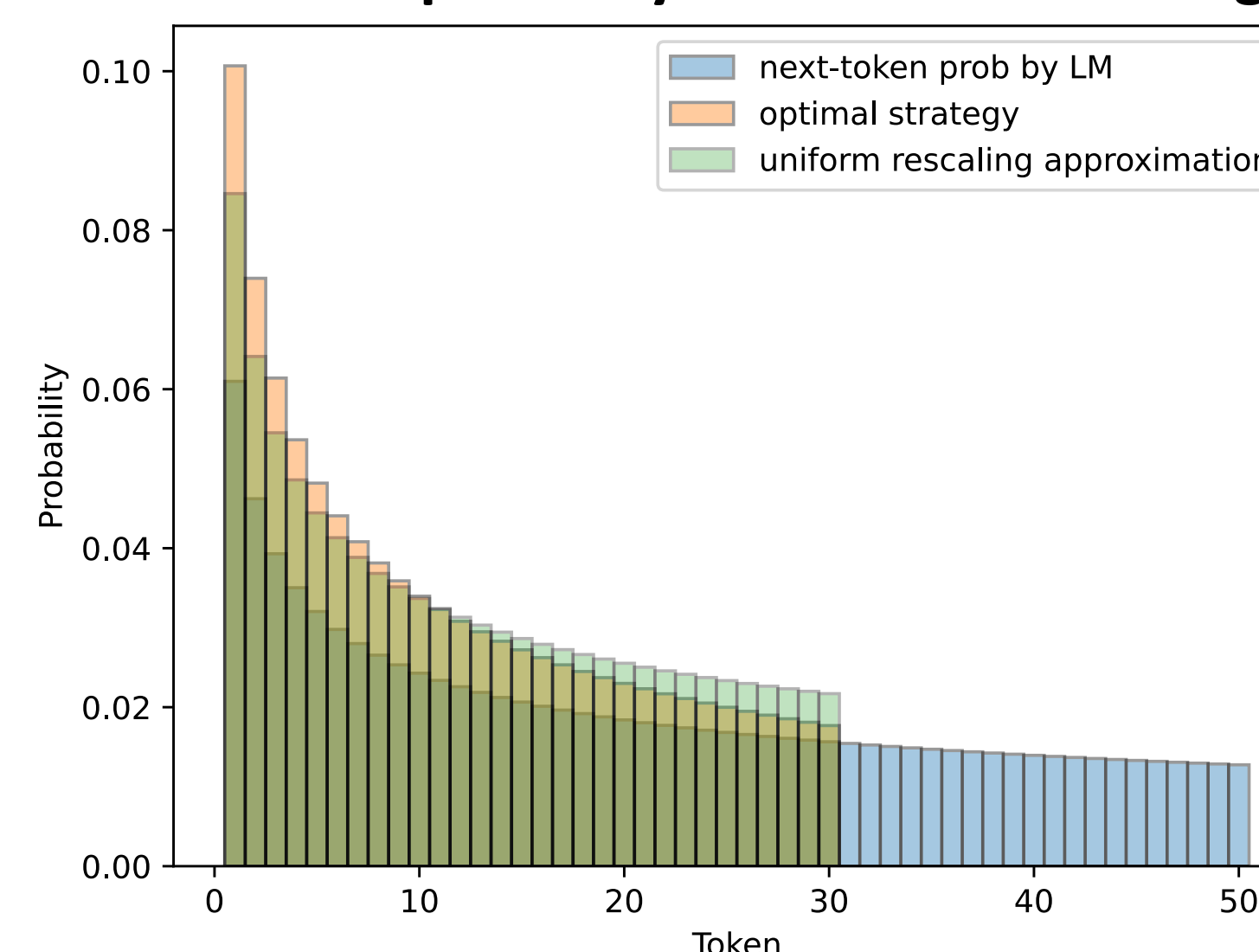
Theorem 4.7. Let $\hat{I} = \max \left\{ I: \sum_{i=1}^{I-1} \frac{f(\hat{p}_i) - f(\hat{p}_I)}{f(\hat{p}_i) - f(\hat{p}_i - \epsilon)} \leq 1 \right\}$. Then

$$q_i^* = c \frac{\epsilon}{f(\hat{p}_i) - f(\hat{p}_i - \epsilon)} \mathbf{1}\{i \leq \hat{I}\},$$

uniform rescaling requires
 $x = 1/f'(x) \Rightarrow f(x) = \log x (+C)$.

where c is the normalizing constant.

Illustration of tail truncation and near-optimality of uniform rescaling



Evaluating optimality in open-ended text generation tasks

